

Causality - Francesco Caratello

Experiment Design:

- #1: Give vaccine to children, compare to prev. year. $\rightarrow$ confirm. bias, bad idea
 - #2: $\rightarrow$ consent $\rightarrow$ test group
 $\rightarrow$ non consent $\rightarrow$ control group } bad Sel. bias
 - #3: Population $\rightarrow$ consent $\rightarrow$ randomly split }
 $\rightarrow$ non consent $\rightarrow$ test group | control group
- randomized trial, RCT $\rightarrow$ gold standard

Causal Inference from observational data

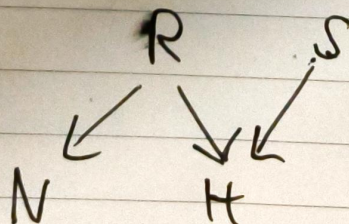
$\rightarrow$ probabilistic inference (causal)

- example:

- $H=1$: Holmes lawn is wet
 - $R=1$: Rain
 - $S=1$: Sprinkles on
- N : Neighbors lawn wet

$$\Rightarrow P(R=1 \text{ or } S=1 | H=1) > P(R=1 \text{ or } S=1)$$

Bayesian Networks:



Parent child
 $x \rightarrow y$

Parents of (i) in Graph

$$\text{Def: } P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{PA}(i))$$

$$\Rightarrow \underbrace{P(N, H, R, S)}_{2^4 - 1} = P(H | R, S) P(N | R) P(R) P(S)$$

We need DAGs for that!
 Lo factorization of joint distribution

DEF: Local Markov Property:

Let $G := (V, E)$ be a DAG, $S \subseteq V$ is arbitrary subset

$$X_S \perp\!\!\!\perp X_{\text{non descendant}(S) \mid PA(S)} \mid PA(S)$$

DEF: Collider Node: Non endpoint node in DAG is a collider node if path contains $\rightarrow i \leftarrow$

DEF: d-separation. A path between nodes i and j is Bloked by a set $S \subseteq V \setminus \{i, j\}$ if

- $\exists$ non-collider $u \in P, u \in S$
- $\exists$ collider $u \in P, u \notin S$ and descendants $(u) \cap S \neq \emptyset$

$\rightarrow u \rightarrow$
 $\leftarrow u \leftarrow$
 $\leftarrow u \rightarrow$

DEF: if all paths between i, j are bloked by S
 ~~$i \perp\!\!\!\perp j \mid S$~~ $i \perp_d j \mid S$

DEF: Global Markov Property: $(\perp_d : d\text{-sep}), (\perp_p : \text{independence})$

P satisfies G, M with respect to DAG G if
 $A \subseteq S, B \subseteq S, A \cap B = \emptyset$ then

$$A \perp_d B \mid S \Rightarrow X_A \perp_p X_B \mid S$$

$$1 \rightarrow 2$$

$$3 \rightarrow 4 \rightarrow 5$$

$$x_1 \perp x_3 \mid x_4 \text{ or } x_5$$

DEFs above are important for Causal Discovery

CAUSAL BAYESIAN NETWORK

vertices, Edges

Let $G = (V, E)$ be a DAG and P be the distribution of X_V . The pair (G, P) is a CAUSAL DAG MODEL (CAUSAL BAY. Net) if

$$\forall W \subset V \quad P(X_V \mid \text{do}(X_W = x'_W)) =$$

$$= \prod_{i \in V \setminus W} P(x_i \mid \text{PA}(i)) \cdot \delta(X_W = x'_W)$$

$$= \begin{cases} \prod_{i \in V \setminus W} P(x_i \mid \text{PA}(i)) & \text{if } X_W = x'_W \\ 0 & \text{otherwise} \end{cases}$$

Example: $P(Y=1 \mid X=1) > P(Y=1 \mid X=0)$

Pass Exam Attend Lecture

$$\Rightarrow P(Y=1 \mid \text{do}(X=1)) > P(Y=1 \mid \text{do}(X=0))$$

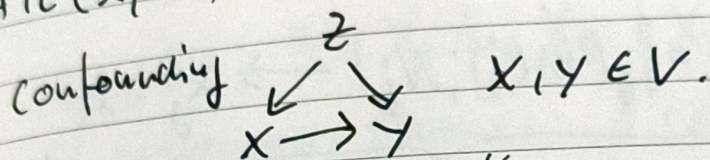
Conditioning: unforced, $\text{do}(\cdot)$: forced

Let X has a causal effect on Y if $\exists a, b$ s.t.

$$P(Y \mid \text{do}(X=a)) \neq P(Y \mid \text{do}(X=b))$$

AVERAGE TREATMENT EFFECTS / CAUSAL EFFECTS

$$ATE(X, X') = E[Y | do(X=X)] - E[Y | do(X=X')]$$



The causal effect of X on Y is confounded if $P(Y|X) \neq P(Y|do(X))$

$$P(Y | do(X=1)) = \sum_z P(Y | X=1, Z=z) \cdot P(Z=z)$$

Structural Equation Models (SEM)

Let $\mathcal{G}$ be a CAUSAL DAG over $X_1, \dots, X_d$.

A SCM/SEM is a collection of structural assignments

$$\{X_j = f_j(PA(j), N_j)\}_{j=1}^d \text{ and } P(N_1, \dots, N_d) = \prod_i P(N_i)$$

PROP: SCM $\rightarrow$ Unique P

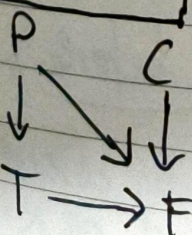
Example: - $P \in \{\text{bus, subway}\}$ (choose transport)

"if bus, always buy ticket, if subway, buy ticket w. p=0.8"

- $T \in \{\text{yes, no}\}$ (buy ticket)

- $C \in \{\text{bus, subway}\}$ (controller in ...)

- $F \in \mathbb{N}$ (Fine to pay)



$$P \sim P_{\text{Bus}} = 0.5 \text{ (Bern(0.5))}$$

$$T \leftarrow \text{bus or } N_T = 1, N_T \sim \text{Bern}(0.8)$$

$$C \sim P_{\text{Bus}} = 0.3, F = 100 \cdot \mathbb{I}(P=C) \mathbb{I}(T=\text{no})$$

$\Rightarrow E[F] \sim$ joint dist., bays. modelling

$$E[F|T=no]$$

INTERVENTION: $x_j = f_j(PA(i), N_i) \rightarrow x_j = x_i$

changing total x_j to x_i HARD, PERFECT INTERV.

Set $N_j = N_i$ do ($x_i = N_i$) with $N_i \sim P$

IMPERFECT INTERVENTION

do ($T=no$) (Never buy a ticket) every bayesian
is SCM but
not vice versa

COUNTERFACTUAL SCM:

$(S, P(N))$ is an SCM. Given observation

x counterfactual SCM: $(S, P(N|X=x))$

exempl. $x = (P=bus, C=bus, T=yes, F=0)$

but what could have happened if ~~the~~ $T=no$?

eg. helpful for policy evaluation

POTENTIAL OUTCOMES: GOLD STANDARD for obs. Data

$Y_i(X=0)$ and $Y_i(X=1)$ are, defined before

the treatment assignment

$$ATE: \frac{1}{n} \sum_i Y_i(X=1) - Y_i(X=0) \quad \left| \begin{array}{l} \text{only observe} \\ x=0 \text{ or } x=1 \end{array} \right.$$

DAY 2 Causality

DEF: GLOBAL MARKOV PROPERTY

$$A \perp_d B \mid S \Rightarrow X_A \perp\!\!\!\perp X_B \mid X_S$$

DEF: ADJUSTMENT FORMULA: A set $Z \subset V \setminus \{i, j\}$ is a valid adjustment set if

$$\underbrace{P(x_i \mid \text{do}(x_j))}_{\text{causal estimand}} = \underbrace{\int_{x_z} P(x_i \mid x_j, x_z) P(x_z) dx_z}_{\text{statistical estimand}}$$

$\begin{array}{c} Z \\ \swarrow \searrow \\ x \rightarrow y \end{array}$ DEF: LOCAL MARKOV: $X_j \perp\!\!\!\perp X_{\text{non descendant}(j)} \mid \text{PA}(j) \mid X_{\text{pa}(j)}$
DEF: MARKOV FACTORIZATION: $P(X_V) = \prod_i P(x_i \mid \text{PA}(i))$

CAUSAL DISCOVERY

$P(X, Y)$ there is an SCM $Y = f_Y(X, N_Y)$
with $X \perp\!\!\!\perp N_Y$... but $x \rightarrow y$ or $y \rightarrow x$???

FAITHFULNESS: $A \perp_d B \mid S \Leftarrow X_A \perp\!\!\!\perp X_B \mid X_S$

$$x_1 \leftarrow \epsilon_1$$

$$x_2 \leftarrow 2x_1 + \epsilon_2$$

$$x_3 \leftarrow 4x_1 - 2x_2 + \epsilon_3$$

$$= \epsilon_2 + \epsilon_3$$

$$x_1 \rightarrow x_2$$

$$\searrow x_3 \swarrow$$

Questions

• $x_1 \perp\!\!\!\perp x_3$: $\checkmark$

• $x_1 \perp_d x_3$: $\times$

cond. independence is not d-separation

Is P FAITHFUL to G ? $\times$

Is P Markov w.r.t. G ? $\checkmark$

DEF: MARKOV EQUIVALENT

Two DAGs G_1, G_2 are MARKOV EQUIVALENT if
 $A \perp_d B \mid S \text{ in } G_1 \iff A \perp_d B \mid S \text{ in } G_2$

PC (Peter & Clarke) Algorithm

STEP 1: Find Skeleton (Graph, no directions)

- START FULL GRAPH (any to any)

for $k = 0, \dots, \# \text{Vertices} - 2$

for i, j adjacent in G

Remove edge if $x_i \perp x_j \mid x_s$

where $x_s \subseteq \text{adjacent}(i, j) \text{ or } \text{adj}(i, j)$
and $|S| = k$

$$i * j \stackrel{\text{Loc. Markov}}{\iff} i \perp_d j \mid \text{PA}(i) \text{ or } \text{PA}(j) \iff$$

$$\iff i \perp_d j \mid \underbrace{\text{Adj}(i) \text{ or } \text{Adj}(j)}_S$$

$$\stackrel{\text{Global Markov}}{\iff} x_i \perp x_j \mid x_s \quad S$$

STEP 2: Find V-Structure

$$- i \rightarrow j \leftarrow u \iff i \not\perp u \mid j$$

STEP 3: Orient all other possible edges

↳ no cycles, no new V-Structure

ML-based approaches (Linear)

$$x \rightarrow y, \quad y = f_{\text{lin}}(x, N_y) = \alpha x + N_y$$

non gaussian noise

LINGAM ALG.

$$X = BX + E, \quad X \in \mathbb{R}^P, \quad B \in \mathbb{R}^{P \times P} \text{ with } \text{diag}(B) = 0$$

($\exists$ permutation π s.t. $B_{\pi\pi}$

$$\text{e.g. } B_{\pi} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \end{bmatrix}$$

$$\text{STEP 1: } (I - B)X = E \Rightarrow X = (I - B)^{-1} E$$

$$A = (I - B)^{-1}$$

Find π of rows s.t. $A_{\pi\pi}^{-1}$ is non-zero on diagonal

observe matrix?

$$A: ICA$$

$\varepsilon_i \perp \varepsilon_j$
Indep. Comp. Anal.
not identifiable if gaussian.

No TEARS

$$X = BX + E \rightarrow \min_{B \text{ is DAG}} \|X - BX\|^2 + \lambda \|B\|_1$$

B is DAG constraint is discrete, hard to optimize, relax by re-writing it continuous!

$\hookrightarrow B \text{ is DAG} \Rightarrow \text{Tr } e^{B \odot B}$ non convex constraint

RESULT: Assume $x \rightarrow y, \quad y = f_{\text{lin}}(x) + \varepsilon$

do linear regression $\hat{f}(x) = \hat{y}, \quad y - \hat{y} = \varepsilon \parallel x$
or ML, Kernel, NNs...

$\Rightarrow$ expensive, need to regress all variables on all other variables

if true then $x \rightarrow y$

Non-Linear Methods

$$x_i = \underbrace{f_i(PA(i))}_{\text{non linear}} + N_i, \quad \underbrace{N_i \sim N(0, \sigma_i)}_{\text{strong assumption}}$$

SCORE-BASED

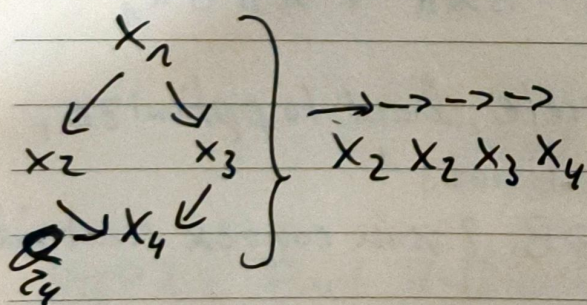
$$\log P(x) = -\frac{1}{2} \sum_i^P \left(\frac{x_i - f_i(PA(i))}{\sigma_i} \right)^2 - \frac{1}{2} \sum_i^P \log 2\pi \sigma_i^2$$

$$\underbrace{\nabla_x \log P(x)}_{S_j(x)} \cdot j = \underbrace{\frac{x_j - f_j(PA(j))}{\sigma_j}}_{\text{only dep. on } j} + \underbrace{\sum_{i \in \text{child } j} \frac{\partial f_i(PA(i))}{\partial x_j} \cdot \frac{x_i - f_i(PA(i))}{\sigma_i^2}}_{\text{only if children}}$$

$$\frac{\partial S_j}{\partial x_j} = \frac{1}{\sigma_j^2} \text{ iff } x_j \text{ is a leaf} \Rightarrow \text{Gives us topological ordering!}$$

Example:

$$S_4 = \frac{x_4 - f_4(PA(4))}{\sigma_4^2}$$



IDENTIFIABILITY MARGIN

$$x_j \rightarrow x_i \quad \uparrow \quad N_i$$

$$\mathbb{E}_{p(x_i)} \left[\left(\frac{\partial^2 f_i(PA(i))}{2\sigma_j^2} \right)^2 \right] \geq C \sigma_i^2 \quad \text{SNR}$$

C is max one can find, like Sign. to Noise Ratio

Var $\frac{\partial S_i}{\partial x_j} \geq C$ if x_j is not a leaf node
 $\Rightarrow$ (weak) finite sample guarantees!

Relaxing Gaussian Noise Assumption: SCORE $\heartsuit$ RESIT

$$S_i(x) = \frac{\partial i}{\partial x_i} \log P(x_i | PA(x_i)) + \sum_u \frac{\partial i}{\partial x_i} f_u(PA_u)$$

$$S = \frac{\partial \ell}{\partial x_i} \log(P(N_i)) , N_i = x_i - f(PA_i) \left\| \frac{\partial i}{\partial x_i} \log(P(x_i | PA_i)) \right\|$$

Score of leaf node is non-lin function of its noise

$$R_i = x_i - \underbrace{E[x_i | x_{\setminus \{i\}}]}_{\text{non-lin. regression}}$$

$$= \begin{cases} N_i & \text{if } x_i \text{ is leaf} \\ N_i - E[N_i | x_{\setminus \{i\}}] & \text{otherwise} \end{cases}$$

everything above from purely observational data

INVARIANT CAUSAL PREDICTIONS

$$x_1 \rightarrow y \rightarrow x_2$$

$$\text{Model I: } E[y | x_1],$$

$$\text{Model II: } E[y | x_2]$$

CASTING CAUSAL DISCOVERY AS CLASSIFICATION PROBLEM!